OSNIAS CLEARING

Decentralised netting with minimal risk for businesses

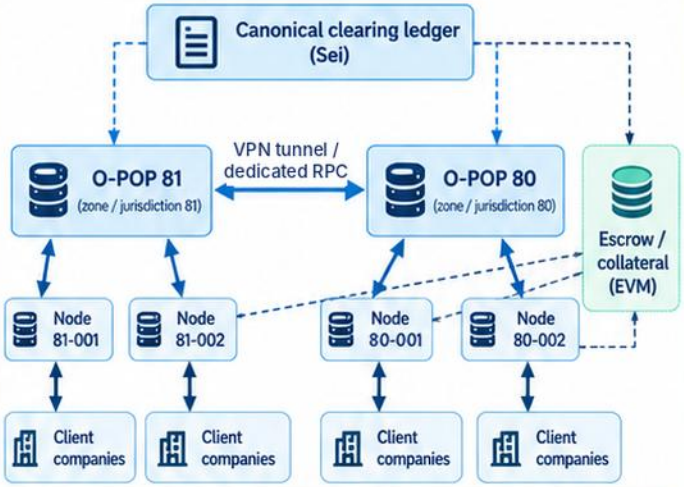
1. Principle

Osnias Clearing is a multilateral netting protocol between companies. It reduces liquidity requirements by offsetting payables and receivables before settlement. The network is decentralised, non-custodial and without mutualisation of funds.



Only net balances are settled.

2. Decentralised network



Nodes operate locally; clearing is coordinated by a canonical ledger, while collateral remains on a separate EVM layer.

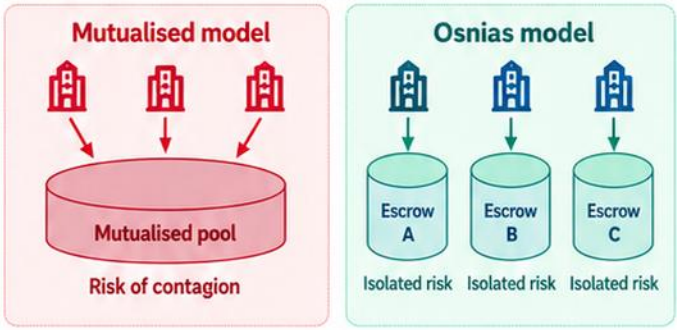
3. Client benefits

- Reduction of liquidity requirements
- Fewer gross payments, more net settlements
- Traceability and auditability of the cycle
- Access to a network of interconnected companies

The denser the network, the lower the treasury requirement.

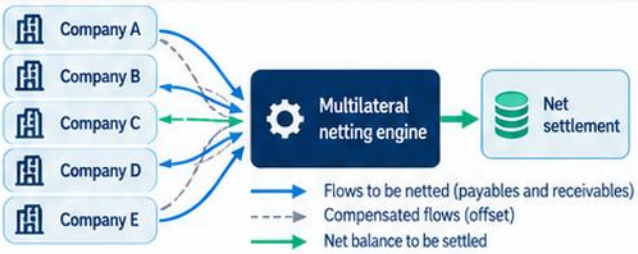
4. Security and risk reduction

- **Non-custodial:** Osnias does not hold client funds
- **No mutualisation:** each user keeps a dedicated escrow
- Separation of clearing / collateral / settlement
- Public solvency oracle and control of the global ratio
- Osnias-ID addressing and strict cycle rules



Risk is brought down to the lowest level at the end-user, as each position remains individualised.

5. Why the network is powerful



Network effect: more compatible counterparties

Compression of flows and collateral efficiency

Scalability by nodes and jurisdictions

The strength of the system comes from multilateral netting: the more participants, the higher the compensation efficiency.

6. Key invariants

Decentralised · Non-custodial · No mutualisation · Reduced risk · Net settlement · Full traceability

Osnias Clearing aims to provide a robust, distributed and low-risk netting infrastructure, focused on the real economy.

OSNIAS CLEARING

Democratising Access to Multilateral Clearing

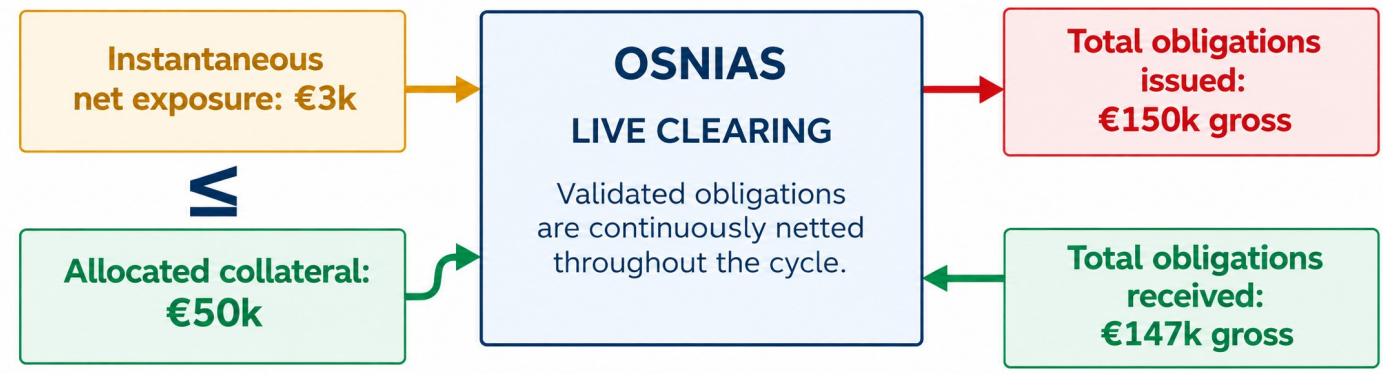
NON-COMMERCIAL EDUCATIONAL DOCUMENT — Executive overview of the architecture and operating model of Osnias Clearing. This document does not constitute a commercial offer, solicitation, or regulatory qualification.

THE PROBLEM

Mid-sized companies must mobilise substantial liquidity to settle gross commercial flows even while they simultaneously hold receivables against other counterparties. This gross circulation of payments structurally increases working-capital requirements, tightens treasury liquidity and ties up capital. One underlying issue is the limited access of corporates to multilateral clearing infrastructure.

The answer: net before settlement

Osnias Clearing organises multilateral clearing within closed cycles. For B2B commercial obligations, the reference cycle is CLEAR 9W: 9 weeks between two settlements, including a 59-day operational window. Users issue, receive and validate obligations; clearing runs live throughout the cycle. Collateral covers only the instantaneous net debit exposure, on a 1:1 basis — never the gross volume of obligations.

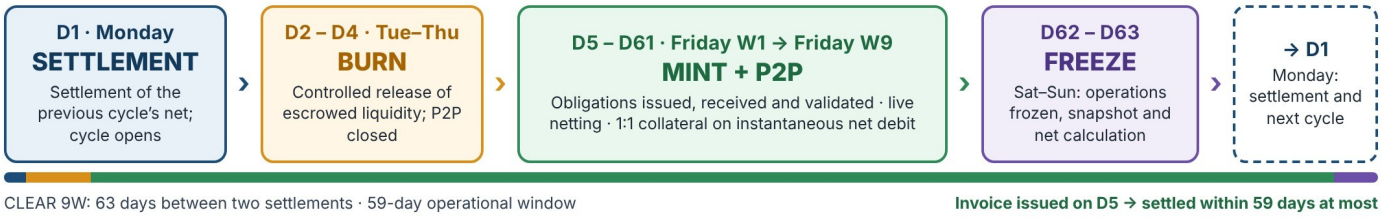


GROSS VOLUME	RISK RULE	EXAMPLE
May be significantly higher than collateral	Instantaneous net debit ≤ collateral	€150k issued / €147k received → €3k net

In the example above, without netting, the company would need to fund €150k from its working capital to pay its suppliers before receiving payment from its own customers.

By posting €50k of collateral in the Euro 9W chamber, the company turns its collateral over three times with counterparties in the same chamber and only has a €3k balance to settle, i.e. 2% of the flow and 6% of the collateral. By entering the clearing process, participants offset reciprocal positions and settle only their residual net balance, without requiring a multiple of that amount in working capital. The liquidity released can be redeployed elsewhere.

Example: the CLEAR 9W cycle (B2B commercial obligations)



D1 (Monday): settlement of the previous cycle and opening of the new cycle.
D2–D4: BURN, partial or full release of liquidity from the escrow account; P2P closed.
D5–D61: MINT + P2P and live clearing.
D62–D63 (weekend): FREEZE and end-of-cycle netting; the following Monday is D1 of the next cycle.

A DECENTRALISED ARCHITECTURE
RISK KEPT AT PARTICIPANT LEVEL

No pooled funds • no leverage • no credit

CLIENT BENEFIT

Lower liquidity and working-capital requirements by settling only the net balance.

• Network access with individually allocated collateral from €10k.

• Gross clearing volume may be materially higher than posted collateral.

• Clearing tools made accessible to a broader range of companies.

NETWORK SECURITY

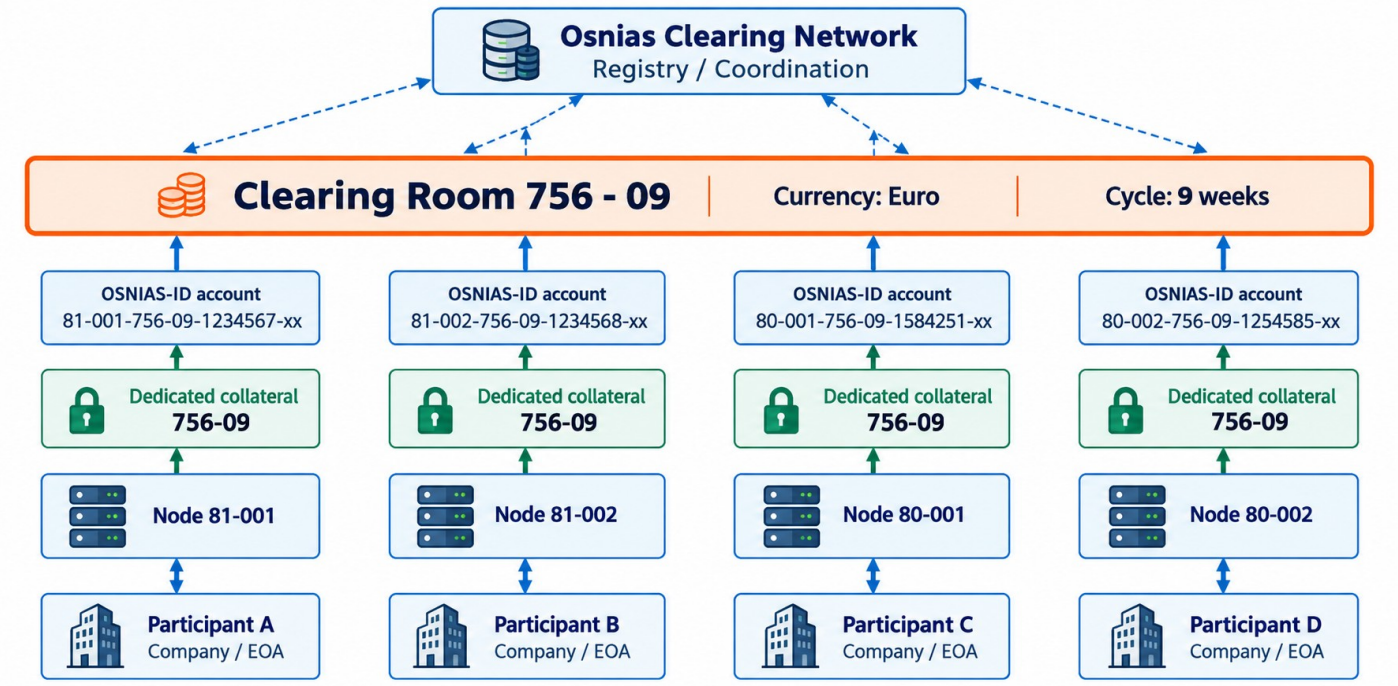
Each instantaneous net debit exposure is covered 1:1 by dedicated collateral.

• No mutualisation: no common pool and no mechanical contagion between users.

• No leverage and no credit: the network does not permit uncollateralised exposure.

• The node operator monitors its users and can block a participant before it creates additional risk.

Controlled exit of a sanctioned participant



SECURITY PRINCIPLE — Each participant is individually segregated and its net exposure is covered 1:1, with no leverage, no credit and no mutualisation of funds. Architecture designed with reference to the PFMI (CPMI-IOSCO).

SECURITY ARCHITECTURE & NETWORK EXPANSION

Strict separation between infrastructure, node operator and end user

Cryptographic separation of functions: 3 addresses, 3 seeds, 3 roles



Osnias Clearing holds no seed phrase and has no ability to move escrowed funds.

OSNIAS CLEARING	NODE OPERATOR	END USER
Infrastructure, cycle rules, netting calculation, ledger and coordination. No onboarding/exit decisions. No control over funds.	Onboarding, monitoring, blocking, exit and operational administration of escrow under its local rules.	Issues, receives and validates obligations; remains linked to its individually allocated collateral and settles its own net balance.

Clearing and collateral: two separate layers

ISO 20022 messages and validated obligations are recorded on Sei, the canonical clearing ledger, accessed through dedicated RPC endpoints via the O-POPs, without a private blockchain network. Only Osnias-IDs appear on it; the mapping to legal identity remains in the node's database. Collateral is held separately on Ethereum in a dedicated smart contract. This separation prevents Osnias Clearing from combining the ledger function with control over participant funds.

Regulatory positioning

Osnias Clearing is a distributed multilateral clearing infrastructure organised as a network of nodes. It does not rely on the interposition of a central counterparty and does not replicate the functional model of a CCP. Osnias does not hold participant funds, does not control escrowed collateral, and does not decide participant onboarding, blocking or exit. The architecture is developed with reference to established principles for risk management, collateralisation, transparency and functional segregation, including the CPMI-IOSCO PFMI. The project is currently in regulatory scoping and testnet phase; no certification, authorisation or regulatory qualification is claimed.

REGULATORY FAQ

Architecture, risk, responsibilities and regulatory scoping

Non-commercial educational document — questions that may be raised by a regulator or institutional partner.

1. Is Osnias Clearing a central counterparty (CCP)?

Osnias is not designed around the functional model of a CCP that interposes itself between every seller and buyer. It organises distributed multilateral clearing among participants and does not perform central novation of obligations. The final regulatory qualification remains subject to scoping with the competent authorities.

2. Why use the term “clearing”?

The term describes the function performed: recording obligations, calculating positions, multilateral netting and determining net balances. If an authority requests different terminology, the preferred wording is “distributed multilateral clearing infrastructure” rather than “clearing house”, which carries a more specific legal meaning.

3. Does Osnias claim PFMI compliance or certification?

No. Osnias does not claim certification, authorisation or formally assessed PFMI compliance. The architecture is designed with reference to relevant CPMI-IOSCO PFMI principles covering risk management, collateral, liquidity, default rules, governance and transparency.

4. What is the project’s current regulatory status?

The project is in development, testnet and preliminary regulatory scoping. No production deployment is presented as authorised, certified or approved. The purpose of the preliminary scoping is to clarify the roles, network model, node operators and settlement framework before production.

The project requires regulatory scoping for node operators, which handle onboarding, identity verification and account monitoring and, where applicable, escrow administration under a contract with the user.

5. Who holds user funds?

By default, the user holds all three keys: the key to its clearing address on Sei, the key to its escrow on Ethereum and the key to its address receiving released funds. Its collateral is individually segregated in a dedicated smart contract. The user may entrust the administration of its escrow to its node operator under a private contract, to which Osnias is not a party. Osnias Clearing only has network-administration access: it holds no keys, provides no custody and cannot move participant funds.

6. Does Osnias hold users’ private keys or seed phrases?

No. Osnias holds none of the three seed phrases associated with the three distinct functions: the clearing 0x address on Sei EVM, the escrow 0x address on Ethereum, and the 0x address receiving released funds.

7. Can Osnias act directly on a user’s clearing account?

No. Any clearing-side action requires authorised access to the O-POP, through VPN, to a dedicated RPC endpoint on Sei, together with the key to the 0x account linked to the Osnias-ID. Osnias does not hold that seed phrase and cannot sign on behalf of the user.

8. Can Osnias act directly on the Ethereum escrow?

No. Any escrow-side action requires the key to the 0x account linked to the Osnias-ID on Ethereum. Osnias holds neither that seed phrase nor the seed phrase for the address receiving released funds. It therefore cannot initiate a movement of funds on behalf of the participant.

9. Who decides participant onboarding, blocking or exit?

The node operator. Osnias provides the infrastructure, common rules and calculation functions, but does not decide user admission, continued participation, blocking or exit.

10. What happens if an obligation would exceed the available collateral?

The transaction fails. The control is enforced by the network's Solidity code: an obligation whose registration would cause the permitted net debit exposure to exceed available collateral is rejected. The rule "instantaneous net debit exposure ≤ collateral" is therefore an execution invariant.

11. Does collateral limit the gross volume of obligations?

No. It limits only the instantaneous net debit exposure during the clearing cycle. A participant may issue and receive unlimited gross volumes during a cycle, provided its instantaneous net debit exposure remains less than or equal to its collateral.

12. Does Osnias permit credit, overdrafts or leverage?

No. The protocol permits neither overdrafts, internal credit nor leverage. There is no mechanism that allows Osnias to grant discretionary exposure above available collateral. This would breach the protocol's invariants.

13. Are funds or losses mutualised?

No. Collateral is individually allocated at participant level. There is no common guarantee fund designed to absorb losses from other participants. Risk is deliberately kept at the lowest level: the individual user.

14. How does the forced exit of a blocked participant work?

A participant placed in blocked status by the node operator can no longer create new exposures. Obligations already registered remain in scope until cycle close and final netting. After settlement, any releasable balance may be withdrawn during the following BURN window in accordance with network rules. The node operator may then change the status from "blocked", which is reversible, to "revoked", which is irreversible and represents the highest sanction level.

15. How is the clearing cycle organised?

For B2B commercial obligations, the CLEAR 9W cycle lasts 9 weeks, i.e. 63 days between two settlements: D1 (Monday) is dedicated to settlement of the previous cycle; D2 to D4 to BURN; D5 (Friday) to D61 to MINT + P2P and live clearing; D62–D63 (weekend) to FREEZE and calculation of the end-of-cycle netting result. The following Monday is D1 of the next cycle. An admitted invoice is therefore settled within 59 days of issuance at most, within the 60-day limit applicable to commercial transactions. The same sequence applies to CLEAR 1W (testnet) and CLEAR 13W (eligible non-commercial obligations, 91 days). Currency and cycle are set when the account is opened and are encoded in the Osnias-ID (Book 4).

16. How are obligations issued and tracked?

Obligations are issued by users through ISO 20022 messaging integrated into the Osnias network. Messages and validated obligations are recorded on Sei, which acts as the canonical clearing ledger, providing a timestamped, verifiable and immutable audit trail.

The canonical ledger is kept on Sei, a public network, and accessed through dedicated RPC endpoints behind the Osnias O-POPs, reachable via VPN, without a private blockchain network (Book 7). Only Osnias-IDs appear in ISO 20022 messages; the purpose field must not contain any identifying data (Rule R-PURPOSE, Book 3).

17. Why separate Sei and Ethereum?

Sei carries the canonical clearing ledger and position records; it is designed for high-speed, low-cost execution. Ethereum provides a highly decentralised, mature and resilient settlement environment. Its transaction costs are higher, but the number of transactions is limited because only collateral-escrow operations are performed there.

This technical and operational separation creates a strict boundary between the escrow chain and the operational clearing chain.

18. How does cross-jurisdiction clearing work?

Obligations are distributed freely by users according to their commercial relationships, within the same chamber (same currency, same cycle). A French participant on node 85-003 may receive an obligation from a German supplier on 79-005 and issue an obligation to a Spanish customer on 74-008. If those obligations are valid and recorded, they enter the netting process; each participant then settles its own net balance.

19. Which law applies to cross-jurisdiction obligations?

Opening the Osnias network in a jurisdiction requires appropriate local regulatory scoping. Each node operator remains responsible for its users and local obligations. The network then processes valid obligations under its common rules; the node operator does not have to requalify each obligation simply because it originates from a different KK jurisdiction code.

20. Can a node operator reject an already validated foreign obligation?

Under the protocol's functional logic, an obligation recorded and validated by the parties enters the netting process under the network's common rules. Access compliance is addressed upstream through jurisdictional scoping and participant onboarding, rather than through discretionary requalification of each obligation during the cycle.

21. Is the clearing token a speculative blockchain innovation?

No. It is an internal technical representation of clearing positions within the Osnias ledger. Internal representations of rights, obligations and balances have long existed in clearing and settlement infrastructures. Blockchain is used here primarily to provide a shared, traceable and immutable ledger.

22. What is the exact role of Osnias Clearing?

Osnias provides decentralised clearing infrastructure: the network's common rules, position calculation, the canonical ledger and traceability.

Osnias is responsible for O-POP nodes, VPN access servers, traceability and data retention for up to 10 years. It does not hold funds, does not hold user keys, does not provide credit and does not decide participant onboarding or exit.

Osnias adapts node-module configuration to each jurisdiction in line with recommendations from the relevant regulators.

Osnias assigns the KK code of each jurisdiction. KK-NNN node numbers are assigned by the competent authority of the territory, or under a procedure placed under its control.

Osnias implements decisions of the association of node operators, which serves as the network's internal governance authority.

23. Why seek preliminary regulatory scoping before production?

Because the project is sufficiently advanced to demonstrate its architecture and controls on testnet while remaining pre-production. Preliminary scoping would allow a mainnet-like testnet configuration to be developed and subjected to security stress testing and audit. This is a fundamental step before submitting a formal application to the relevant regulatory authority with an audited testnet.

This is the rigorous path before mainnet deployment.

24. How is VAT handled?

Osnias nets net-of-VAT amounts only. The VAT on each invoice remains calculated, invoiced and reported in companies' ERP systems under the ordinary rules: Osnias does not calculate, collect or remit any tax. The netting statement (camt.054) allows the ERP to reconcile each invoice. The interaction with e-invoicing and the ViDA package is described in the public note "ViDA 2030" (www.osnias-clearing.com/en/vida2030/), which also puts forward, prospectively and for discussion, a reverse-charge option on netted entries.

The answers above describe the project's functional architecture at its current stage.

They do not constitute a definitive legal qualification or a claim of regulatory authorisation.